RenalGlyco AI

Software Architecture & System Requirements Specification (SRS/SAD)
Target Conditions: Type 2 Diabetes Mellitus (T2D) & Chronic Kidney Disease (CKD)

1. Executive Summary & Project Identification

RenalGlyco AI is an open-architecture clinical decision support platform designed to identify early-stage risk of Type 2 Diabetes Mellitus (T2D) and Chronic Kidney Disease (CKD). Metabolic dysregulation and renal impairment frequently co-exist; elevated blood glucose levels cause long-term microvascular damage to kidney nephrons, while compromised renal function disrupts metabolic equilibrium.

The system ingests routine non-invasive clinical parameters—such as blood panels, electrolyte measurements, and urinalysis results—and processes them through automated machine learning pipelines to compute dual disease risk scores. It incorporates SHapley Additive exPlanations (SHAP) to deliver explainable AI (XAI) feature impact metrics to clinicians.

2. Detailed Problem Statement

2.1 Medical Context & Clinical Pain Points

Chronic metabolic and renal diseases develop silently over prolonged periods. Clinicians face three structural challenges during early patient screening:

Static Threshold Reliance: Standard clinical diagnostic workflows evaluate parameters against static reference intervals (e.g., Fasting Plasma Glucose >= 126 mg/dL or Serum Creatinine > 1.2 mg/dL). This approach overlooks subtle non-linear interactions across multiple borderline parameters.

Asynchronous Disease Tracking: Diabetes and renal damage are frequently managed in clinical silos. Early renal hyperfiltration and mild nephron loss often go undetected until significant decline in Estimated Glomerular Filtration Rate (eGFR) has occurred.

The AI 'Black-Box' Adoption Barrier: Machine learning classifiers are often rejected by medical practitioners due to a lack of interpretability. Without an audit trail explaining which biomarkers elevated a patient's risk calculation, predictions cannot be translated into targeted clinical interventions.

2.2 System Objectives

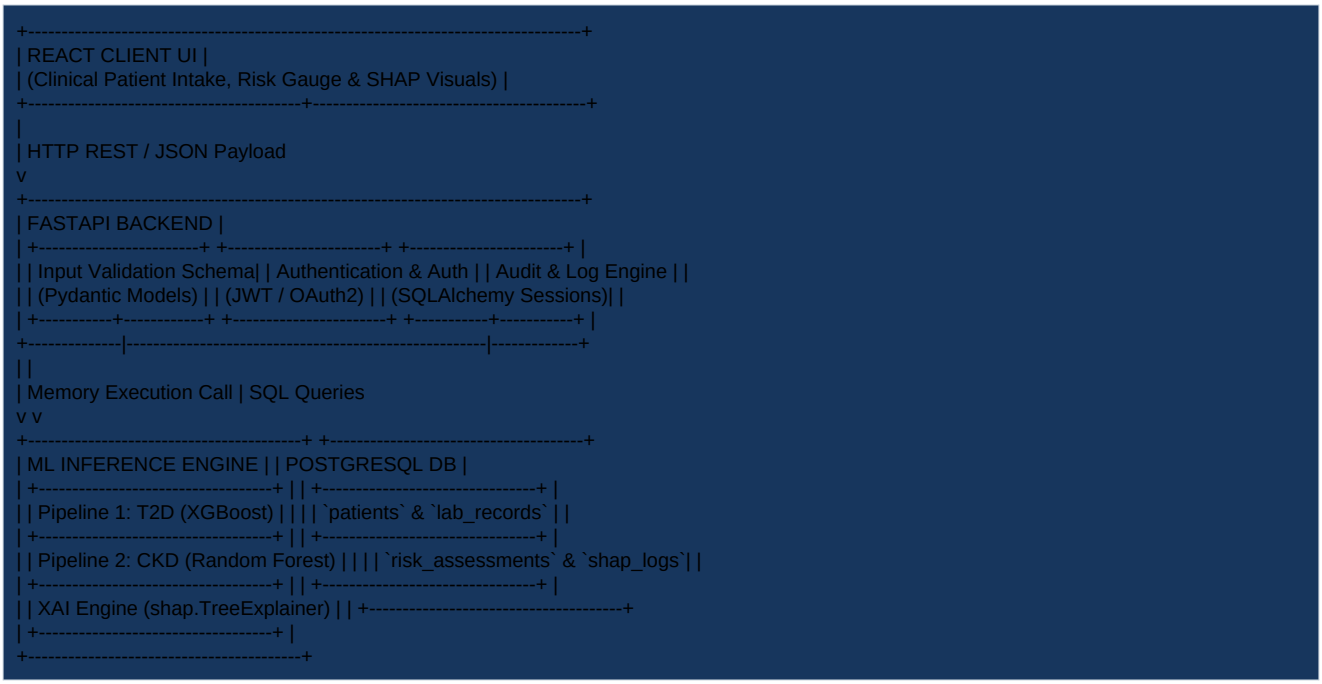
Predictive Accuracy: Deliver an F1-Score >= 0.86 and ROC-AUC >= 0.88 across independent validation cohorts without using cardiovascular data.

Explainability: Compute exact per-patient feature attribution scores to rank top biological risk drivers.

Interoperability & Usability: Provide a lightweight REST API gateway paired with an intuitive clinical interface for instant risk evaluation and historical trajectory tracking.

3. Proposed Solution Architecture

RenalGlyco AI uses a decoupled micro-architecture comprising an asynchronous FastAPI backend, a machine learning inference engine, a PostgreSQL persistence store, and a React frontend dashboard.



4. Datasets & Feature Engineering Specifications

The engine is trained and validated on two benchmark clinical datasets from the UCI Machine Learning Repository.

4.1 Dataset A: Type 2 Diabetes Dataset (Pima / NHANES Cohort)

Sample Count: 768 patient profiles | Target Feature: Outcome (0 = Non-Diabetic, 1 = Diabetic)

Feature Column	Measurement Unit	Value Range	Imputation Strategy
pregnancies	Count	0 - 17	Median Strategy
glucose	mg/dL	44.0 - 199.0	KNN (k=5)
blood_pressure	mm Hg	24.0 - 122.0	KNN (k=5)
skin_thickness	mm	7.0 - 99.0	Median Strategy
insulin	mu U/mL	14.0 - 846.0	KNN (k=5)
bmi	kg/m^2	18.2 - 67.1	KNN (k=5)
diabetes_pedigree	Linear Score	0.078 - 2.42	Mean Strategy
age	Years	21 - 81	Direct Validation

4.2 Dataset B: Chronic Kidney Disease (CKD) Dataset

Sample Count: 400 patient records | Target Feature: classification (ckd = 1, notckd = 0)

Feature Column	Measurement Unit	Clinical Description	Data Type
age	Years	Patient Age	Continuous
bp	mm Hg	Blood Pressure	Continuous
sg	Specific Gravity	Urine Density (1.005 - 1.025)	Categorical / Ordinal
al	0 - 5 Scale	Albumin in Urine	Categorical / Ordinal
su	0 - 5 Scale	Sugar in Urine	Categorical / Ordinal
bgr	mg/dL	Blood Glucose Random	Continuous

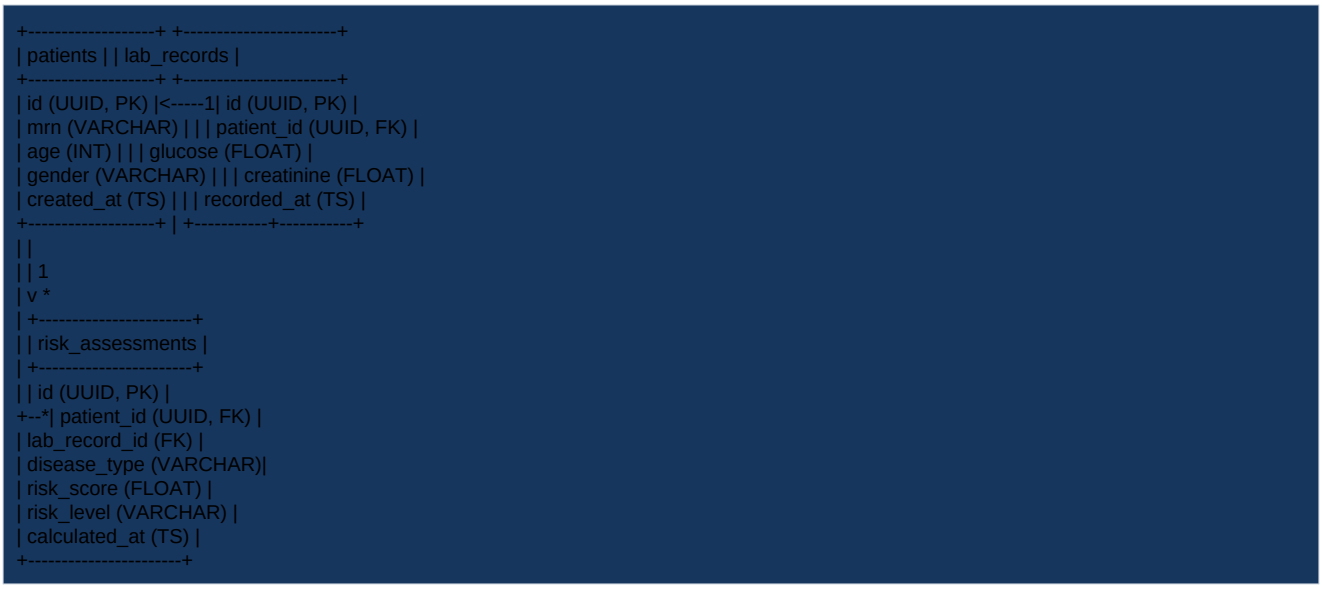
Feature Column	Measurement Unit	Clinical Description	Data Type
bu	mg/dL	Blood Urea	Continuous
sc	mg/dL	Serum Creatinine	Continuous
sod	mEq/L	Blood Sodium	Continuous
pot	mEq/L	Blood Potassium	Continuous
hemo	g/dL	Hemoglobin Level	Continuous
pcv	%	Packed Cell Volume	Continuous
wbcc	cells/cumm	White Blood Cell Count	Continuous

5. Technology Stack Specifications

Technology Layer	Selected Tool / Framework	Justification & Architectural Role
Language	Python 3.11	High performance, strict type annotations, native ML library compatibility.
Data Processing	NumPy, Pandas, Scikit-Learn	Vectorized tabular operations, scaling, cross-validated pipeline construction.
Machine Learning	XGBoost, Random Forest	Optimized gradient boosting and decision tree ensembles for tabular clinical data.
Explainability	SHAP (shap.TreeExplainer)	Fast calculation of exact Shapley values for tree-based ensemble classifiers.
Backend Framework	FastAPI + Uvicorn	Asynchronous execution, auto-generated OpenAPI schemas, strict JSON validation.
Database & ORM	PostgreSQL 15 + SQLAlchemy 2.0	Transactional relational storage, strict foreign key integrity, schema migrations via Alembic.
Frontend UI	React 18 + TailwindCSS	Responsive, component-driven clinical interface with client-side state handling.
Data Visualization	Chart.js / Recharts	Client-side rendering of feature attribution bar graphs and risk progress gauges.
Containerization	Docker + Docker Compose	Isolated environment build packaging for API services and database nodes.

6. Database Schema & API Specifications

6.1 Relational Database Layout



6.2 API Endpoint Response Payload

`POST /api/v1/predict/multi-risk`


```
{
"status": "success",
"assessment_id": "a1b2c3d4-e5f6-7a8b-9c0d-1e2f3a4b5c6d",
"predictions": {
"diabetes": {
"risk_probability": 0.68,
"risk_category": "Moderate",
"top_risk_drivers": [
{"feature": "Glucose", "shap_impact": 0.28},
{"feature": "BMI", "shap_impact": 0.19}
]
},
"chronic_kidney_disease": {
"risk_probability": 0.84,
"risk_category": "High",
"top_risk_drivers": [
{"feature": "Serum Creatinine", "shap_impact": 0.38},
{"feature": "Urine Albumin", "shap_impact": 0.24}
]
}
},
"timestamp": "2026-10-26T21:19:00Z"
}
```


7. Execution & Implementation Roadmap

Phase	Milestone Name	Timeline	Key Tasks & Technical Deliverables
Phase 1	Foundation & Data Pipeline	Weeks 1 - 2	Environment setup, Git workflow initialization, dataset collection, and data cleaning pipelines.
Phase 2	Model Engineering & XAI	Weeks 3 - 4	Train XGBoost and Random Forest classifiers; integrate SHAP explainer modules; export serialized models (.joblib).
Phase 3	Backend & Database API	Weeks 5 - 6	Develop FastAPI endpoints, configure Pydantic schemas, connect PostgreSQL via SQLAlchemy models.
Phase 4	Frontend Dashboard Build	Week 7	Construct React user interfaces, clinical input forms, risk gauges, and SHAP visual charts.

Phase	Milestone Name	Timeline	Key Tasks & Technical Deliverables
Phase 5	Integration & Deployment	Week 8	Run full test suites (pytest), package application with Docker, deploy to hosting infrastructure.